

Research Topics for Text Mining (TMG301)

From Course Projects to Research-Oriented Projects

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From Course Project to Research Project

Implementation-oriented project

Dataset → Algorithm → Train → Accuracy

Research-oriented project

Research Question → Baselines → Proposed Comparison / Improvement → Controlled Experiments → Analysis → Conclusion

The key difference is not necessarily a new algorithm.

A good student research project can investigate:

- When does one method outperform another?
- Which component actually contributes to performance?
- How robust is the model under noisy or limited data?
- What are the trade-offs between accuracy and computational cost?
- Why does the model fail on particular linguistic phenomena?

Expected Research Structure

Each team should define:

- ➊ **Problem:** What NLP/Text Mining problem are we studying?
- ➋ **Research Questions (RQs):** What do we want to discover?
- ➌ **Dataset:** Which data can answer the RQs?
- ➍ **Baselines:** What existing approaches should be compared?
- ➎ **Investigation:** What will we improve, vary, or analyze?
- ➏ **Experiments:** How will the hypotheses be tested?
- ➐ **Evaluation:** Which metrics are appropriate?
- ➑ **Analysis:** Why do the results occur?
- ➒ **Output:** Paper + code + model + demo/visualization.

20 Research Topics – Overview

Text Processing & Classification

- ➊ Vietnamese Text Normalization
- ➋ N-Gram Language Modeling
- ➌ Spam Detection
- ➍ Sentiment Analysis
- ➎ Aspect-Based Sentiment Analysis
- ➏ Hate Speech Detection
- ➐ Word Embedding Evaluation
- ➑ Semantic Textual Similarity
- ➒ Fake News Detection
- ➓ Efficient BERT Fine-Tuning

Advanced NLP

- ➑ Neural Machine Translation
- ➒ Low-Resource Machine Translation
- ➓ FAQ Chatbot
- ➑ Intent & Slot Extraction
- ➒ Vietnamese Syntactic Parsing
- ➓ Relation Extraction
- ➑ Event Extraction
- ➒ Temporal Information Extraction
- ➓ Word Sense Disambiguation
- ➑ Emotion Detection

Topic 1 – Vietnamese Text Normalization

Research Title

Evaluating Text Normalization Strategies for Noisy Vietnamese Social Media Text

Problem

Vietnamese social media text contains abbreviations, slang, missing diacritics, repeated characters, emojis, spelling errors, and non-standard Unicode representations.

Objective

- Develop a Vietnamese text-normalization pipeline.
- Quantify how normalization affects downstream NLP tasks.
- Identify which types of normalization contribute most.

Possible Data

- Vietnamese social-media comments.
- UIT-VSFC or other public Vietnamese sentiment datasets.
- A manually created noisy → normalized test set.

Topic 1 – Research Design

Research Questions

- RQ1: Does normalization improve Vietnamese text classification?
- RQ2: Which normalization operation contributes most?
- RQ3: Can aggressive normalization remove useful sentiment signals?

Baselines

Raw Text, Unicode Normalization, Rule-based Normalization

Proposed Investigation

- Regex + dictionary + edit distance.
- Slang and abbreviation normalization.
- Repeated-character reduction.
- Compare rule-based and contextual normalization.

Experiments

- Raw vs. normalized text.
- Ablation: remove one normalization component at a time.
- Evaluate downstream sentiment classification.

Metrics: CER, normalization accuracy, Accuracy, Macro-F1.

Topic 2 – N-Gram Language Modeling

Research Title

A Comparative Study of Smoothing Strategies for Vietnamese N-Gram Language Models

Problem

N-gram language models suffer from sparse observations and unseen word sequences, particularly for morphologically and lexically diverse text.

Objective

- Investigate how N-gram order affects language modeling.
- Compare smoothing methods under different training-data sizes.
- Study the relationship between vocabulary size and OOV words.

Data

- Vietnamese Wikipedia.
- Vietnamese news corpus.
- Training subsets: 10%, 25%, 50%, 100%.

Topic 2 – Research Design

Research Questions

- RQ1: Which N-gram order provides the best perplexity?
- RQ2: Which smoothing method is most effective under limited data?
- RQ3: How does vocabulary size affect OOV rate and perplexity?

Baselines

Unigram, Bigram, Trigram

Methods

- Maximum Likelihood Estimation.
- Laplace/Add- k smoothing.
- Good-Turing.
- Interpolation / Kneser-Ney.

Experiment Matrix

$N \in \{1, 2, 3, 4, 5\} \times \text{Smoothing} \times \text{Training Size}$

Metrics: Perplexity, OOV rate, model size, inference time.

Output: Vietnamese N-gram LM + experimental benchmark.

Topic 3 – Vietnamese Spam Detection

Research Title

Classical Machine Learning versus Neural Models for Vietnamese Spam Detection

Objective

- Automatically classify spam and legitimate messages.
- Investigate whether complex models are necessary for this task.
- Analyze linguistic indicators of Vietnamese spam.

Data

- Vietnamese SMS/email spam datasets.
- Public comments/messages where legally and ethically appropriate.

Features

- Bag-of-Words / TF-IDF.
- Character N-grams.
- Word embeddings.

Topic 3 – Research Design

Research Questions

- RQ1: Are character N-grams robust to obfuscated spam?
- RQ2: Do neural models significantly outperform TF-IDF?
- RQ3: Which types of spam are hardest to detect?

Baselines

- Multinomial Naïve Bayes.
- Logistic Regression.
- Linear SVM.

Advanced Comparison

- BiLSTM.
- PhoBERT.

Experiments

- Word TF-IDF vs. character TF-IDF.
- Classical ML vs. neural models.
- Robustness against intentional character substitutions.

Metrics: Precision, Recall, F1, Macro-F1.

Topic 4 – Vietnamese Sentiment Analysis

Research Title

From Naïve Bayes to BERT: Revisiting Vietnamese Sentiment Classification

Research Questions

- How much improvement is obtained from contextual representations?
- Does BERT still dominate when training data are limited?
- Which linguistic phenomena cause classification errors?

Data

- UIT-VSFC.
- Vietnamese e-commerce reviews.

Baselines

NB → Logistic Regression → BiLSTM

Advanced Model

PhoBERT

Topic 4 – Experiments and Outputs

Experiments

- 1 TF-IDF + NB/LR.
- 2 Word2Vec/FastText + BiLSTM.
- 3 PhoBERT fine-tuning.
- 4 Train using 10%, 25%, 50%, and 100% of data.

Error Analysis

- Negation.
- Mixed sentiment.
- Slang.
- Sarcasm.
- Domain-specific expressions.

Metrics

Accuracy, Precision, Recall, Macro-F1

Research Output

- Vietnamese sentiment benchmark.
- Trained classifier.

Topic 5 – Aspect-Based Sentiment Analysis

Research Title

Aspect-Based Sentiment Analysis for Vietnamese E-Commerce Reviews

Problem

Document-level sentiment cannot explain *what* users like or dislike.

Example

“Camera đẹp nhưng pin yếu và giao hàng quá chậm.”

Camera : +, *Battery* : –, *Delivery* : –

Research Questions

- How accurately can aspects be extracted from Vietnamese reviews?
- Does joint modeling outperform separate aspect/sentiment models?
- How does performance vary across product domains?

Topic 5 – Research Design

Baselines

- TF-IDF + classifier.
- BiLSTM.
- BiLSTM-CRF for aspect extraction.

Advanced Comparison

- PhoBERT token classification.
- PhoBERT sentiment classifier.
- Optional joint aspect-sentiment model.

Data

- Vietnamese e-commerce reviews.
- VLSP-style ABSA data if available.

Experiments

- Aspect extraction.
- Aspect sentiment classification.
- Cross-domain evaluation.

Metrics: Aspect-F1, Sentiment Macro-F1.

Topic 6 – Hate Speech Detection

Research Title

Robust Vietnamese Hate Speech and Offensive Language Detection

Research Questions

- Can models distinguish hate speech from general offensive language?
- How robust are models to spelling variation and obfuscation?

Dataset

- ViHSD or comparable public Vietnamese datasets.

Baselines

TF-IDF + SVM $\rightarrow$ BiLSTM $\rightarrow$ PhoBERT

Improvement

- Class weighting.
- Data augmentation.
- Character-aware preprocessing.

Metrics: Macro-F1, per-class F1, confusion matrix.

Output: classifier + robustness/error analysis.

Topic 7 – Word Embedding Evaluation

Research Title

Static versus Contextual Representations for Vietnamese NLP

Research Questions

- Do contextual embeddings always outperform static embeddings?
- How well do embeddings capture Vietnamese semantic relationships?
- How does embedding choice affect downstream classification?

Models

TF-IDF $\rightarrow$ Word2Vec $\rightarrow$ FastText $\rightarrow$ PhoBERT

Experiments

- Word similarity.
- Nearest-neighbor analysis.
- Text classification.
- Visualization with PCA/t-SNE.

Metrics: correlation, F1, computational cost.

Output: embedding benchmark + visualization tool.

Topic 8 – Semantic Textual Similarity

Research Title

Semantic Textual Similarity for Vietnamese using Sentence Representations

Research Question

Can semantic embeddings identify sentences with similar meanings even when they share few words?

Baselines *Jaccard* $\rightarrow$ *TF-IDF* + *Cosine* $\rightarrow$ *Word2Vec*

Advanced Methods

- PhoBERT sentence representations.
- Sentence-BERT style models.

Experiments

- Semantic similarity correlation.
- Paraphrase retrieval.
- Hard-negative evaluation.

Metrics *Pearson*, *Spearman*, *Recall@K*, *MRR*

Output: semantic search engine + benchmark.

Topic 9 – Fake News Detection

Research Title

Vietnamese Fake News Detection using Linguistic and Contextual Features

Research Questions

- Are linguistic features sufficient to detect misinformation?
- What does PhoBERT learn beyond TF-IDF?
- Does a model trained on one topic generalize to another?

Baselines TF-IDF + Logistic Regression, TF-IDF + SVM., BiLSTM.

Advanced Model PhoBERT

Experiments

- In-domain test.
- Cross-topic test.
- Feature ablation.

Metrics: Macro-F1, ROC-AUC.

Output: detection prototype + generalization analysis.

Topic 10 – Efficient BERT Fine-Tuning

Research Title

Accuracy–Efficiency Trade-offs in Fine-Tuning Vietnamese Language Models

Research Questions

- Is full fine-tuning necessary for small datasets?
- Can parameter-efficient methods achieve comparable accuracy?
- What is the trade-off between accuracy, memory, and training time?

Methods

- Frozen PhoBERT + classifier.
- Full fine-tuning.
- Adapter/LoRA-based fine-tuning.

Experiments

10%, 25%, 50%, 100% training data

Measure:

F1, Trainable Parameters, GPU Memory, Training Time

Output: efficient fine-tuning benchmark + reusable code.

Topic 11 – Vietnamese–English Machine Translation

Research Title

From Seq2Seq to Transformer: Vietnamese–English Neural Machine Translation

Research Questions

- How much does attention improve Seq2Seq translation?
- How much improvement comes from Transformer architectures?
- What types of Vietnamese sentences remain difficult to translate?

Baselines

Seq2Seq $\rightarrow$ Seq2Seq + Attention $\rightarrow$ Transformer

Data

- PhoMT.
- IWSLT Vietnamese–English.

Metrics

BLEU, chrF, Human Evaluation

Output: NMT system + translation error taxonomy.

Topic 12 – Low-Resource Machine Translation

Research Title

Data Augmentation for Low-Resource Vietnamese–English Machine Translation

Research Questions

- How rapidly does MT quality degrade as parallel data decrease?
- Can augmentation compensate for missing parallel data?
- Which augmentation strategy is most useful?

Baseline *Transformer_{Original}*

Proposed Comparisons

- Back-translation.
- Word/subword noise.
- Sentence filtering.
- Synthetic parallel data.

Experiment $Data \in \{10\%, 25\%, 50\%, 100\%\}$ Compare: Original vs. Original+Augmentation

Metrics: BLEU, chrF, robustness. **Output:** low-resource MT benchmark + augmentation pipeline.

Topic 13 – Vietnamese FAQ Chatbot

Research Title

Retrieval-Based Vietnamese FAQ Chatbot using Semantic Search

Research Questions

- Does semantic retrieval outperform lexical retrieval?
- How sensitive is retrieval to paraphrased questions?
- Can hybrid lexical-semantic retrieval improve reliability?

Baselines

TF-IDF $\rightarrow$ BM25 $\rightarrow$ Dense Retrieval

Improved Approach

$$Score = \alpha Score_{BM25} + (1 - \alpha) Score_{Semantic}$$

Metrics

Recall@K, MRR, Hit@K

Output: Web chatbot + FAQ knowledge base + retrieval evaluation.

Topic 14 – Intent Classification and Slot Filling

Research Title

Joint Intent Detection and Slot Filling for Vietnamese Task-Oriented Dialogue

Example

“Đặt vé máy bay từ Đà Nẵng đi Hà Nội ngày mai.”

Intent:

BookFlight

Slots:

From = DaNang, To = Hanoi, Date = Tomorrow

Research Questions

- Does joint learning improve intent and slot prediction?
- How well do models generalize to unseen utterance forms?

Baselines

- SVM intent classifier.
- BiLSTM.
- BiLSTM-CRF.

Advanced: PhoBERT-based joint model.

Metrics: Intent Accuracy, Slot-F1, Sentence Accuracy.

Output: reusable NLU module for Vietnamese chatbot.

Topic 15 – Vietnamese Syntactic Parsing

Research Title

Evaluating Syntactic Parsing Approaches for Vietnamese Text

Research Questions

- Which Vietnamese syntactic structures are most difficult?
- How much do pretrained representations improve parsing?
- How sensitive is parsing to sentence length?

Data Vietnamese Treebank/dependency treebank.

Methods Traditional/transition-based baseline, Neural dependency parser, Pretrained contextual representations.

Experiments

- Performance vs. sentence length.
- Performance vs. dependency type.

Metrics: UAS, LAS.

Output: parsing benchmark + dependency visualization.

Topic 16 – Relation Extraction

Research Title

Relation Extraction from Vietnamese News using Contextual Language Models

Example

“FPT University được thành lập bởi Tập đoàn FPT.”

(FPT University, founded_by, FPT)

Research Questions

- Do contextual representations outperform pattern-based methods?
- How does entity distance affect relation extraction?
- Can extracted relations construct useful knowledge graphs?

Baselines *Rules* → *TF-IDF + Classifier* → *BiLSTM*

Advanced Model: PhoBERT relation classifier.

Metrics: Precision, Recall, Macro-F1.

Output: relation extractor + small Vietnamese knowledge graph.

Topic 17 – Event Extraction

Research Title

Automatic Event Extraction from Vietnamese News Articles

Goal

Extract: $Event = \{Trigger, Arguments, Time, Location\}$

Research Questions

- Which event types can be reliably extracted?
- Does contextual modeling improve event-trigger detection?
- What errors occur when multiple events appear in one sentence?

Baselines

- Keyword/rule-based event detection.
- BiLSTM sequence labeling.

Advanced Method

- PhoBERT token classification.

Metrics: Trigger-F1, Argument-F1.

Output: event extraction pipeline + event visualization.

Topic 18 – Temporal Information Extraction

Research Title

Extracting and Normalizing Temporal Expressions from Vietnamese News

Example

“Ba ngày trước, công ty công bố sản phẩm mới.”

“Ba ngày trước” → *DATE* – 3*D*

Research Questions

- How well can rule-based methods handle Vietnamese temporal expressions?
- When are contextual models necessary?
- Can extracted temporal information construct event timelines?

Baselines

Regex → *CRF* → *BERT*

Metrics: Precision, Recall, F1, normalization accuracy.

Output: temporal extractor + automatic timeline generator.

Topic 19 – Word Sense Disambiguation

Research Title

Contextual Word Sense Disambiguation using Lexical Knowledge and Transformer Embeddings

Research Questions

- How effective are knowledge-based WSD methods?
- Do contextual embeddings capture different word senses?
- Does additional lexical knowledge improve contextual WSD?

Baselines *Most Frequent Sense* → *Lesk* → *WordNet Similarity*

Advanced Method *BERT/PhoBERT Contextual Representation*

Experiments

- Accuracy by ambiguity level.
- Accuracy by word frequency.
- Knowledge-based vs. contextual approaches.

Output: WSD model + semantic case-study analysis.

Topic 20 – Vietnamese Emotion Detection

Research Title

Beyond Positive and Negative: Fine-Grained Vietnamese Emotion Detection

Emotion Classes

Joy, Sadness, Anger, Fear, Surprise, Disgust, ...

Research Questions

- Which emotions are easiest/hardest to distinguish?
- Do affect lexicons complement contextual embeddings?
- How does class imbalance influence emotion recognition?

Baselines *Lexicon* $\rightarrow$ *TF-IDF* + *LR* $\rightarrow$ *BiLSTM*

Advanced Model *PhoBERT*

Possible Improvement

PhoBERT + *Affect Lexicon Features*

Metrics: Macro-F1, per-class F1.

Output: emotion classifier + emotion-analysis dashboard.

Minimum Experimental Requirements

Each project should include at least:

Three Levels of Models

Simple Baseline $\rightarrow$ *ML/DL Baseline* $\rightarrow$ *Modern/Improved Method*

Example:

TF-IDF + LR $\rightarrow$ *BiLSTM* $\rightarrow$ *PhoBERT*

And at least:

- 2 clearly defined Research Questions.
- 3 compared methods/configurations.
- 1 controlled experiment or ablation study.
- Quantitative evaluation.
- Qualitative/error analysis.

Do Not Stop at Accuracy

Students should answer:

Performance

- Which model performs best?
- Is the difference significant?
- On which classes?

Robustness

- Noisy input?
- Limited data?
- Domain shift?

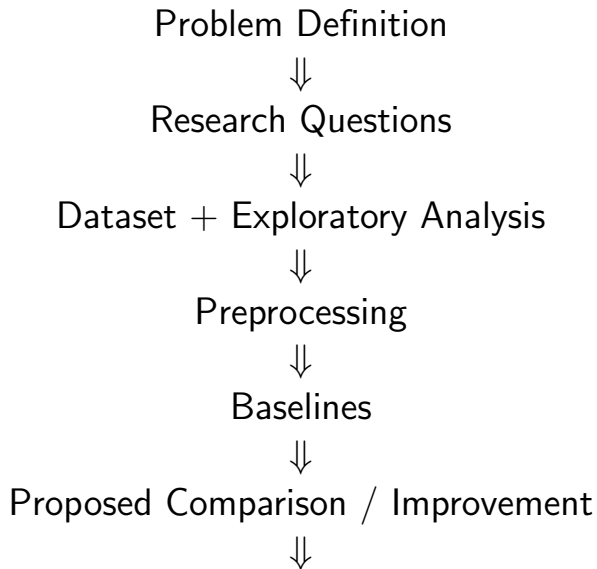
Efficiency

- Training time?
- Inference time?
- Model size?

Explainability

- Why does the model fail?
- Which linguistic cases are difficult?
- What patterns were learned?

Recommended Experimental Pipeline



Expected Research Outputs

Each team should submit:

① Research Report

- Approximately 6–8 pages.
- IEEE/ACM-style structure.

② Source Code

- Reproducible experiments.
- README and requirements.

③ Dataset / Processed Dataset

- Data-processing scripts.
- Data statistics.

④ Trained Model

- Saved checkpoints where appropriate.

⑤ Demo / Visualization

- Web demo, dashboard, or interactive notebook.

Suggested Research Paper Structure

- ① Introduction
 - Problem
 - Motivation
 - Research Questions
 - Contributions
- ② Related Work
- ③ Dataset and Problem Formulation
- ④ Methodology
- ⑤ Experimental Setup
- ⑥ Results and Discussion
- ⑦ Error / Ablation Analysis
- ⑧ Conclusion

Example: Turning a Project into Research

Weak Project Question

“Can we use PhoBERT for sentiment analysis?”

Example: Turning a Project into Research

Weak Project Question

“Can we use PhoBERT for sentiment analysis?”

Better Research Questions

- RQ1: How much does PhoBERT improve over TF-IDF and BiLSTM?
- RQ2: Does this advantage remain when only 10–25% of training data are available?
- RQ3: Which linguistic phenomena benefit most from contextual representations?
- RQ4: What accuracy–efficiency trade-off does PhoBERT introduce?

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Research Mindset

Do not only ask: **“Does the model work?”**

Ask: **“Why, when, and under what conditions does it work?”**

Recommended Team Workflow

Phase 1 – Research Design

Problem → Literature → RQs

Phase 2 – Data

Collection → Cleaning → EDA

Phase 3 – Baselines

Simple ML → Deep Learning

Phase 4 – Research Investigation

Improvement / Comparison / Ablation

Phase 5 – Evaluation

Metrics → Error Analysis → Discussion

Phase 6 – Communication

Paper + Code + Demo + Presentation

What Makes a Good Topic?

A strong project should satisfy four criteria:

1. Researchable

The problem can be expressed through clear Research Questions.

2. Measurable

The questions can be answered through experiments and quantitative metrics.

3. Feasible

The dataset, computing resources, and implementation are achievable within the course.

4. Explainable

Students can explain not only *what* happened but also *why* it happened.

The goal is not to build the most complicated model.

The goal is to conduct a

small but complete research study.

Question → Evidence → Analysis → Conclusion

Thank You

Choose a problem that makes you curious.

Then design experiments that can answer it.